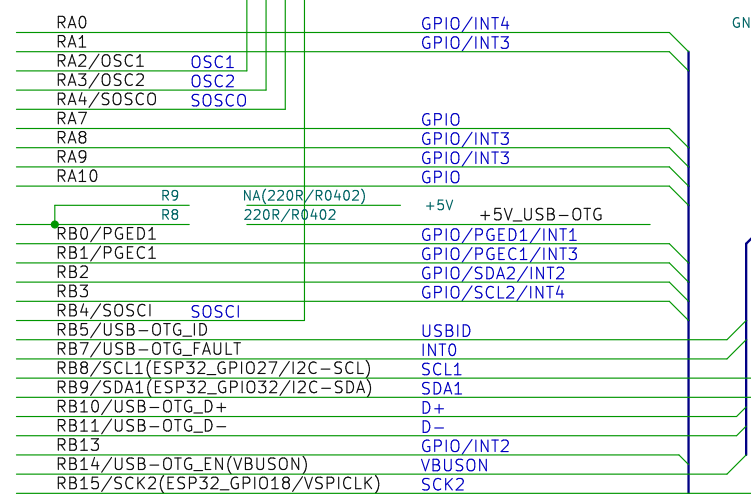
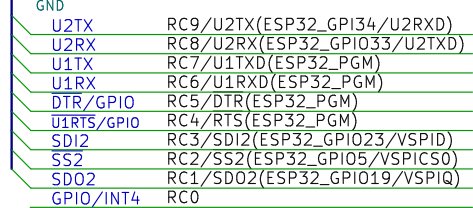


Power Supply



Attention!!!
+5V_Only!!!

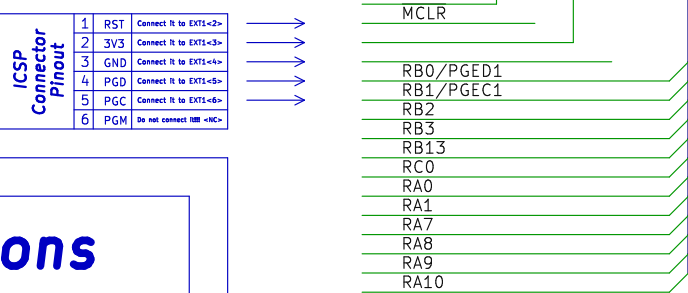
PIC32MX270F256DT-1/PT



The following PIC32MX270F256DT-1/PT registers must be configured as follows:

1. DEVCFG6's bits ICSEL<1:0> to be set to: 1 = PGEC1/PGED1 pair is used;
2. DEVCFG6's bit FUSBIDIO to be set to: 1 = USBID pin is controlled by the USB module;
3. IEOC's bit <3> to be set to: 1, i.e. INTO - External Interrupt is enabled;
4. DEVCFG3's bit FVBUSONIO to be set to: 1 = VBUSON pin is controlled by the USB module;
5. RPC1R<3:0> to be set to: 0100 = SD02, i.e. RC1 = SD02;
6. RPC2R<3:0> to be set to: 0100 = SS2, i.e. RC2 = SS2;
7. SD12R<3:0> to be set to: 0111 = RPC3, i.e. RC3 = SD12;
8. RPC4R<3:0> to be set to: 0001 = U1RT5, i.e. RC4 = U1RT5 or use a GPIO function for RC4;
9. Use GPIO function for RC5/DTR!
10. U1RXR<3:0> to be set to: 0011 = RPC6, i.e. RC6 = U1RX;
11. U1TXR<3:0> to be set to: 0001 = U1TX, i.e. RC7 = U1TX;
12. U2RXR<3:0> to be set to: 0110 = RPC8, i.e. RC8 = U2RX;
13. RPC9R<3:0> to be set to: 0010 = U2TX, i.e. RC9 = U2TX;
14. Set up all other PIC32MX270F256DT-1/PT registers so that to be enabled and correctly configured all other used peripherals!

The components related to both crystal oscillators are consistent to the recommended in the PIC32MX270F256DT-I/PT's datasheets!



Bootstrapping Pins Informations

Software Selectable Pins		
Interface	Signal	Pin
EMAC	EMAC_MDC_out EMAC_MDIO_in EMAC_RS_out EMAC_COL_out	Any GPIO
I2C	I2CEXT_SCL_in I2CEXT_SDA_in I2CEXT_SCL_out I2CEXT_SDA_out I2CEXT_SCL_out I2CEXT_SDA_out	Any GPIO
General Purpose SPI	HSPICLK_in/out HSPID_in/out HSPICSSB_in/out HSPICSSL_out HSPICSS2_out VSPICLK_in/out VSPID_in/out VSPICLK_in/out VSPICSS1_out VSPICSS2_out	Any GPIO

For more information refer to [esp_wdtm_37_dedicated_pin.pdf](#).

Internal Bootstrapping Resistors	
MTDI/GPIO12:	Pull-Down
GPI00:	Pull-Up
GPI02:	Pull-Down
GPI04:	Pull-Down
MTDO/GPIO15:	Pull-Up
GPI05:	Pull-Up

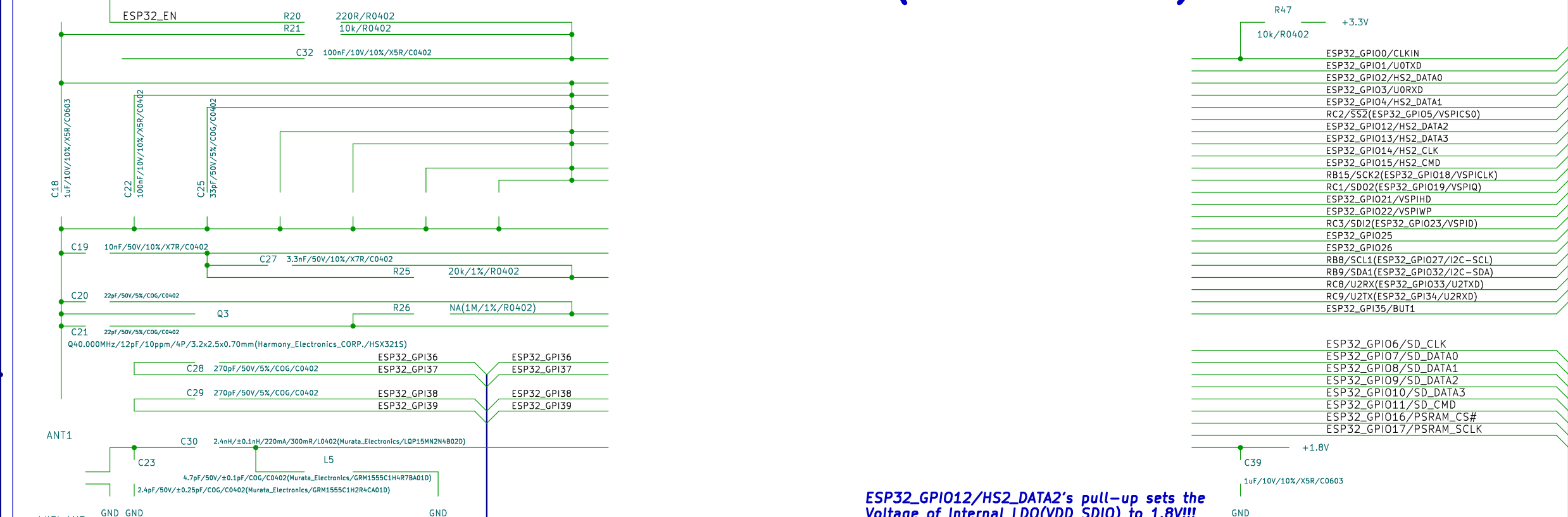
Voltage of Internal LDO(VDD,SDIO)			Bootstrapping Pins Settings	
Pin	Default	3.3V 1.8V		
GPIO12/MTDI	Pull-Down	0 1		
Boot Mode				
Pin	Default	SPI Flash Boot Download Boot		
GPIO0	Pull-Up	1 0		
GPIO2	Pull-Down	Don't-Care 0		
Debugging Log on UOTXD During Booting				
Pin	Default	UOTXD Toggling	UOTXD Silent	
GPIO15/MTDQ	Pull-Up	1 0	0 0	
Timing of SDIO Slave				
Pin	Default	Failing-edge:Input Raising-edge:Output	Rising-edge:input Falling-edge:Output	Rising-edge:input Rising-edge:Output
GPIO15/MTDQ	Pull-Up	0 0	1 0	1 1

Extension Connectors

Notes:
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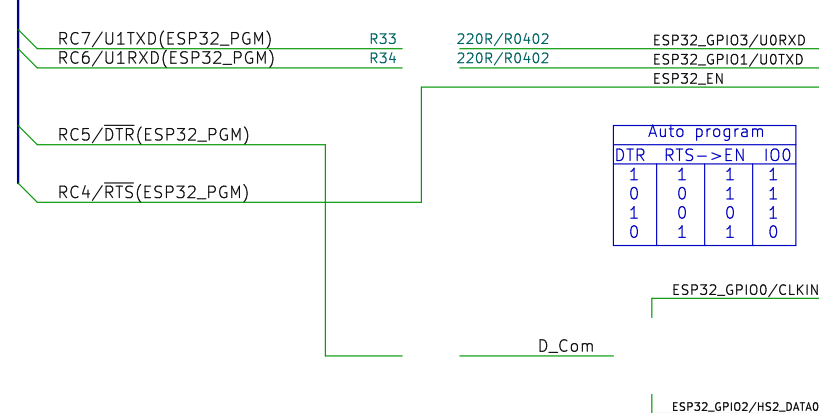
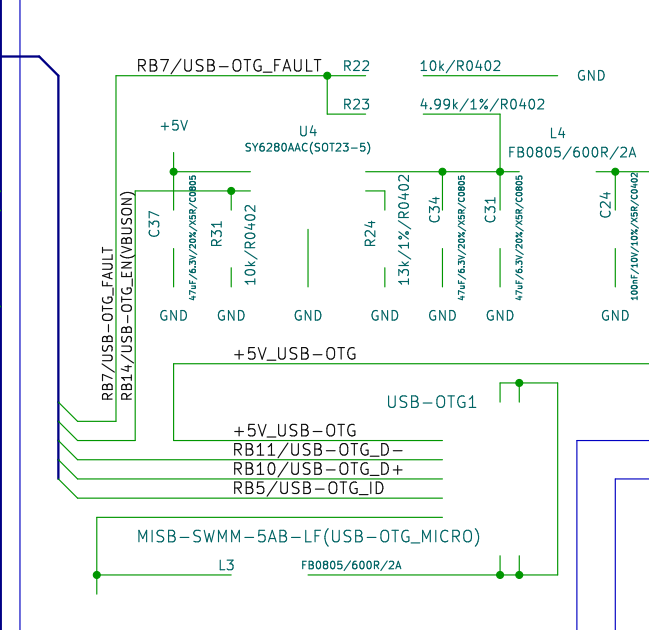
1. According to the datasheet: VSPi Supports Standard SPI, Dual SPI, and Quad SPI that can be connected to the external flash and SRAM!!!

ESP32-D0WDQ6(QFN48_6x6mm)

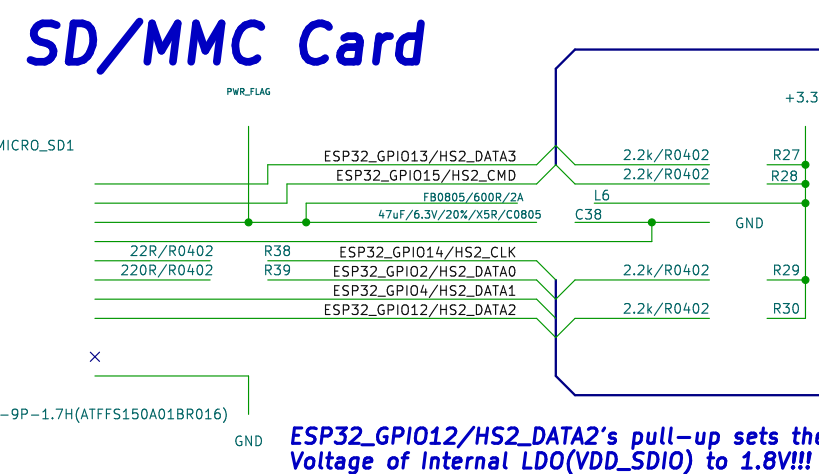


ESP32_GPI012/HS2_DATA2's pull-up sets the Voltage of Internal LDO(VDD_SPI0) to 1.8V!!!

USB-OTG



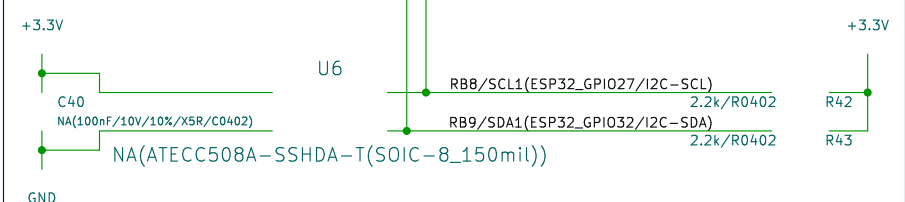
ESP32 programming circuit



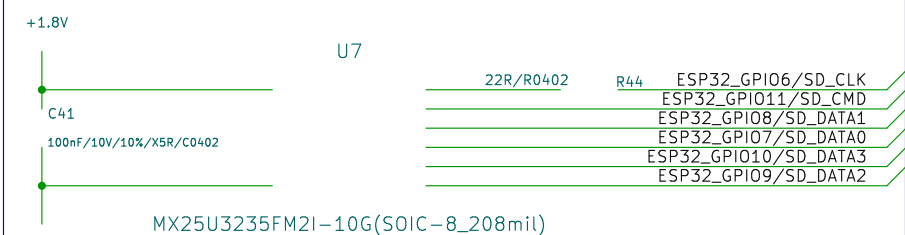
ESP32_GPI012/HS2_DATA2's pull-up sets the Voltage of Internal LDO(VDD_SPI0) to 1.8V!!!

Crypto Authentication Device

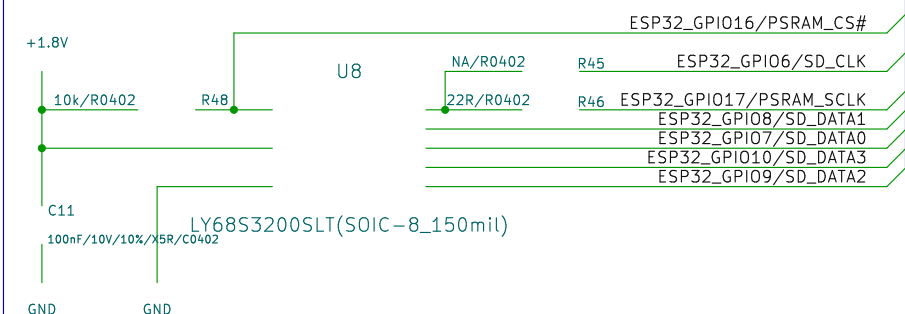
I2C Address: 0xC0



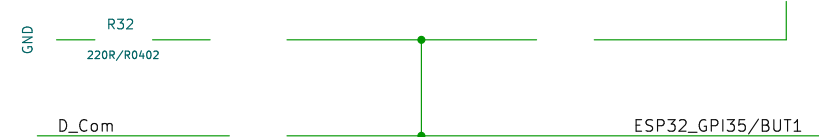
32M Bits Serial MULTI I/O FLASH MEMORY



32M Bits Serial Pseudo-SRAM with SPI and QPI



User Button



Fiducials



<https://www.olimex.com>
OLIMEX LTD.

Sheet: /
File: working.kicad_sch
Title: ESP32-PRO



ware

Size: A2	Date: 2018-07-30
KiCad E.D.A. eeschema 7.0.6	

Rev: B1
Id: 1/1